

AI SOFTWARE DEVELOPMENT INTERNSHIP

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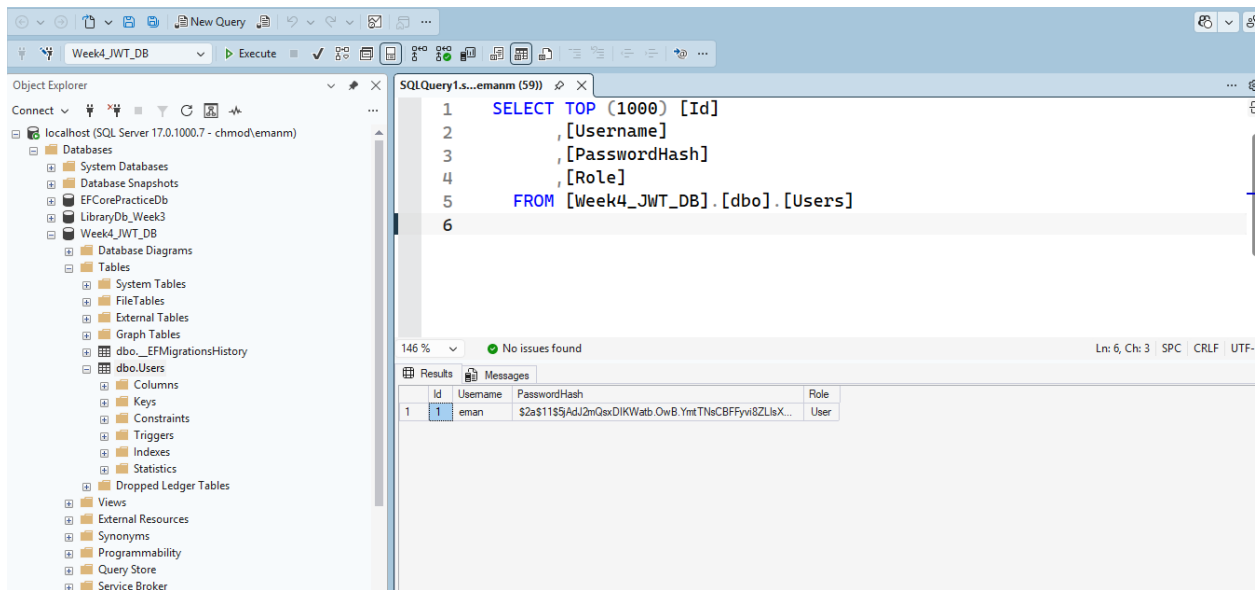
Week 4 — Detailed Task Sheet JWT Authentication (.NET + Angular) • FastAPI
Foundations LLM APIs In Depth • Prompt Engineering • Git — Level Up

PART A — JWT AUTHENTICATION IN ASP.NET CORE (REAL IMPLEMENTATION)

Practice Tasks

1. Register a user through Swagger and confirm the password is stored as a long hash, not plain text, when you look at the row in SQL Server.

Output:



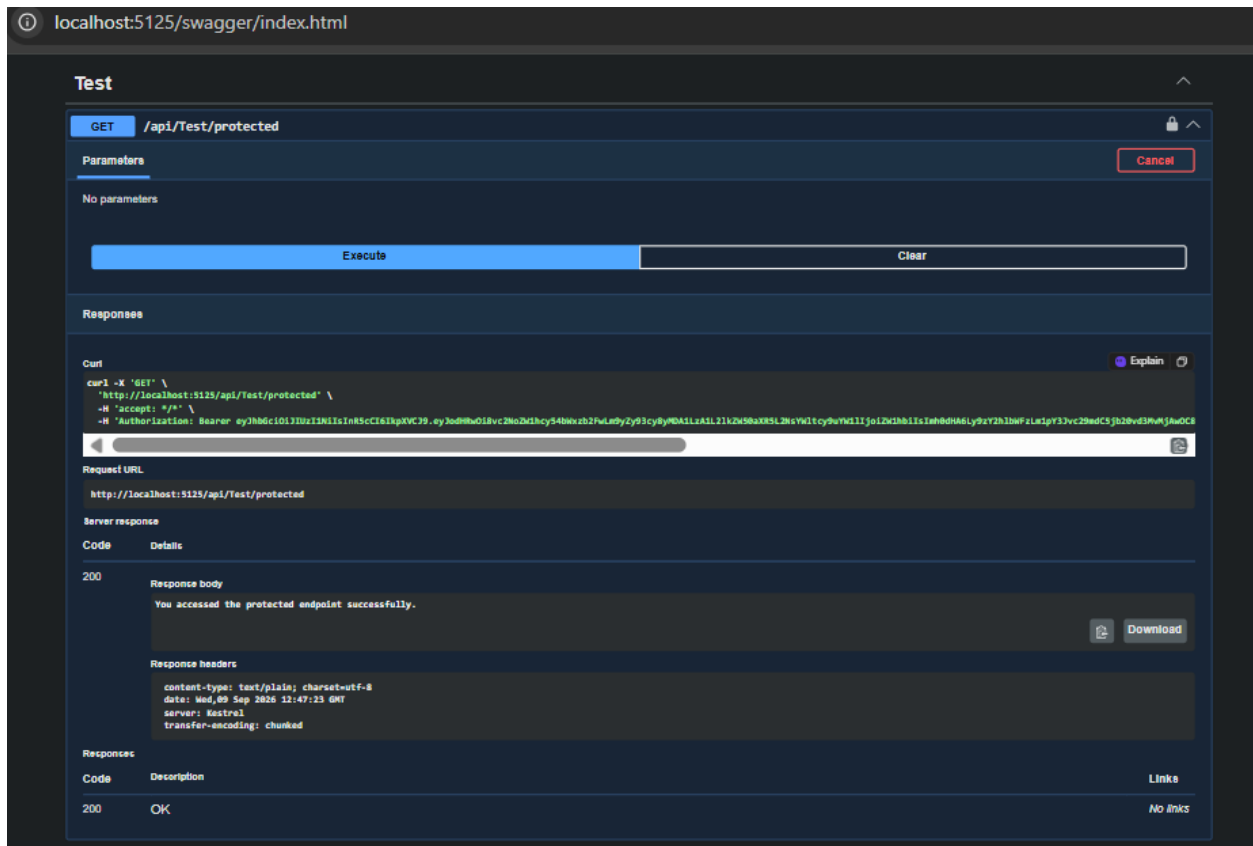
The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'Week4_JWT_DB'. The central pane shows a SQL query executed in the 'SQLQuery1.s...emanm (59)' window. The query is:

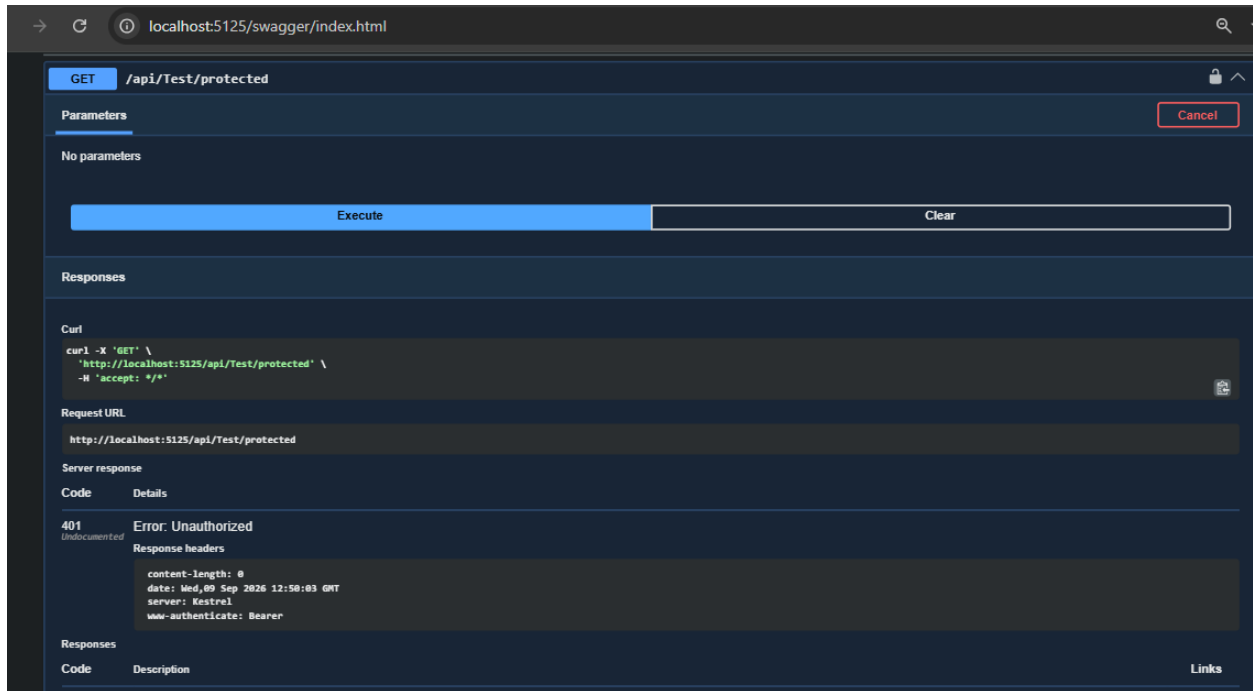
```
1 SELECT TOP (1000) [Id]
2     , [Username]
3     , [PasswordHash]
4     , [Role]
5 FROM [Week4_JWT_DB].[dbo].[Users]
6
```

The Results pane at the bottom shows the output of the query, which is a single row with the following data:

Id	Username	PasswordHash	Role
1	eman	\$2a\$11\$5\$AdJ2mQaxDIKWab:OwB.YmtTNaCBFFyr8ZLbX...	User

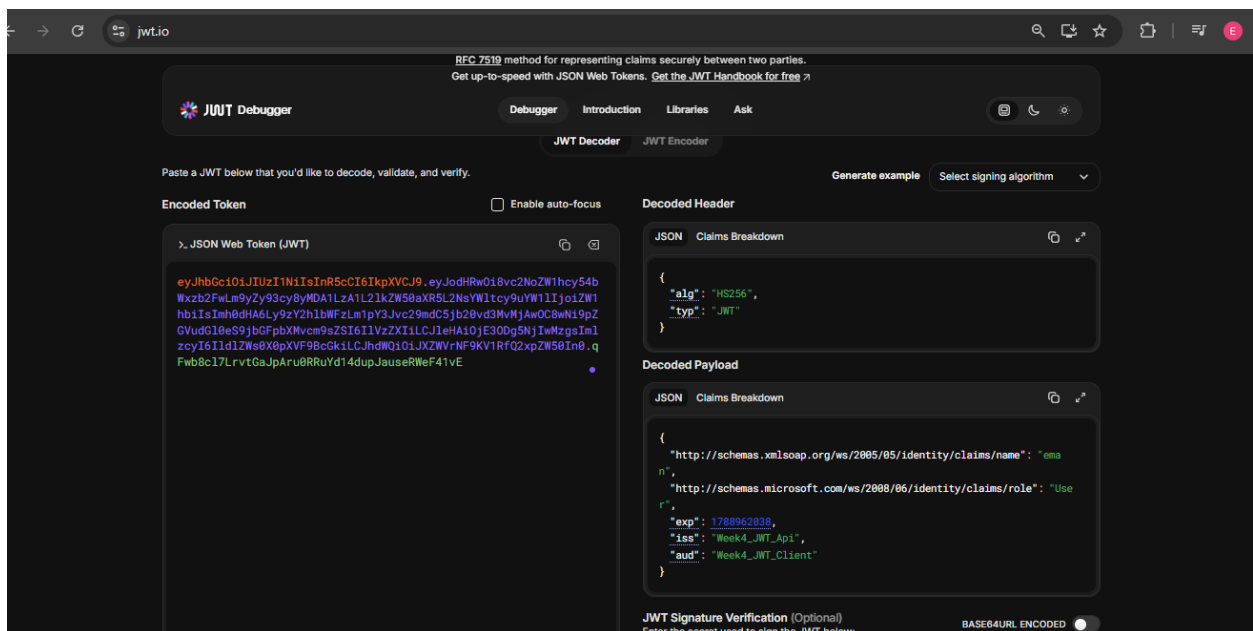
- Output:**





3. Decode your own generated token at jwt.io (paste only the token, never a real production secret) and confirm the claims and expiry look correct.

Output:



4. Add a second user with role Admin and restrict the DELETE endpoint to that role only; confirm a normal user gets 403 Forbidden.

Output:

